5Sense — Technical Flowchart Architecture

Software Flow • Code Execution Pipeline • IBM LinuxONE (s390x)

IBM Z Datathon 2026
Live: nickel-highlight-hide-units.trycloudflare.com
Repo: github.com/Ronquoris/5SeNes.git

FLOWCHART 1: SYSTEM TOPOLOGY FLOW

USER DEVICE (PWA)
- Camera Video (30 FPS)
- YOLOv8 WASM (<10ms)
- Touchpad Gestures
- SpeechQueue Engine
- 3D Binaural Earcons

→ HTTPS

CLOUDFLARE TUNNEL
- Outbound Daemon
- 0 Inbound Open Ports
- Zero-Trust Gateway
- SSL Termination

→ s390x

IBM LINUXONE (S390X)
- Native Node.js 22 LTS
- Fastify REST (:4000)
- In-Memory Hazard/SOS
- Telum II Ready (27ms)
- systemd Supervisor

→ HTTP

GEOSPATIAL APIS
- OSRM Foot Routing
- Nominatim Geocoding
- Overpass POI Search

FLOWCHART 2: CODE EXECUTION FLOWCHART (NAVIGATEPAGE.TSX)

Touch Event on Screen (PointerDown / TouchStart)

TAP (<240MS) • VISION	HOLD (>240MS) • ROUTING	3-TAP / SHAKE • SOS
handlePadPressEnd() Calls handleSimpleDetection()	**handlePadPressStart()** Timer > 240ms triggers push-to-talk	**handleEmergencySos()** Triggered by 3 taps or useShakeDetection()
YoloDetector.detectTick() ONNX WASM tensor scan on video frame	**speechQueue.setMicActive(true)** Mutes all background audio in 0ms	**playSosEmergencyAlarm()** Alternating siren + SOS vibration pulse
estimateDistance(bbox) Computes distance in metres	**startHoldVoiceRecognition()** Web Speech API captures destination	**POST /api/sos/start** Sends GPS to LinuxONE Fastify service
[Distance < 2.5m?] **YES:** playSpatialHazardTone() + speak("Caution...") **NO:** playClearPathChime() + speak("Path clear")	**POST /api/geocode/search** LinuxONE resolves multi-destinations	**Session Storage & Broadcast** Creates tracking token + SMS alerts
	POST /api/route LinuxONE OSRM computes foot route	**Reverse Geocoded Link** Trusted contacts open live tracking map
	Mount VoiceCoPilot.tsx Starts turn-by-turn walking guidance	

5Sense — Navigation & Speech Queue Flowcharts

Voice Co-Pilot Walking Radar • Speech Collision Prevention Logic

IBM LinuxONE Engine
3D Spatial Earcons

FLOWCHART 3: VOICE CO-PILOT FLOW (VOICECOPILOT.TSX)

navigator.geolocation.watchPosition()

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processPosition(coords)
Computes distance: haversineMeters(pos, step.location)

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[Evaluate Distance to Next Turn Corner]

Distance ≤ 20 Metres (Preparation)
- playSpatialTurnCue(dir) → pans chime to turning ear
- speak("In 20 metres, take right onto [Street]", NAVIGATION)
- Single 50ms haptic pulse

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Distance ≤ 8 Metres (Prompt)
- Waypoint ping in turning ear
- speak("In a few steps, take right", NAVIGATION)
- Double 40ms haptic pulse

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Distance ≤ 2 Metres (Corner Execution)
- playSpatialTurnCue(dir) → urgent dual chime
- speak("Take right now onto [Street]", NAVIGATION)
- Advances step: setStepIndex(stepIndex + 1)
- Announces next: "Then, in 45m, take left"

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Compass Veering Detection (DeviceOrientation)
headingDifference(compassHeading, expectedBearing) > 35°
→ Triggers: speak("Veering right. Turn slightly left.")

FLOWCHART 4: SPEECHQUEUE LOGIC (SPEECHQUEUE.TS)

speechQueue.speak(text, priority)

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[Is Microphone Active? (isMicActive == true)]

YES (User Speaking):
Reject or buffer in FIFO queue.
Zero background speech plays.
Mic never records app sound.

NO (Mic Idle):
Proceed to Priority Rank Evaluation.

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[Priority Hierarchy Check]
CRITICAL (0) > NAVIGATION (1) > INFO (2)

Case A: Higher Priority Arrives (CRITICAL)
- Immediate preemption: cancel() running speech in 0ms
- speakNow(CRITICAL): Hazard alert plays instantly

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Case B: Same or Lower Priority Arrives
- Enqueued in sorted FIFO array: queue.push(item)
- Waits for onUtteranceEnd(finishedId) callback

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Watchdog & Chrome Unfreeze Loop
- Universal watchdog auto-advances queue if browser pauses
- Set<SpeechSynthesisUtterance> prevents GC deletion
- 2500ms heartbeat invokes speechSynthesis.resume()

FLOWCHART 5: CODE MODULE INTERCONNECTION

```
NavigatePage.tsx ──> YoloDetector.ts ──> estimateDistance.ts ──> SpeechQueue.ts (CRITICAL Obstacle Alert)
NavigatePage.tsx ──> speechQueue.setMicActive(true) ──> Fastify Backend (/api/geocode, /api/route) ──> VoiceCoPilot.tsx
VoiceCoPilot.tsx ──> playSpatialTurnCue() (StereoPannerNode) ──> Web Audio Earcons (Left/Right Ear) + Haptics
NavigatePage.tsx ──> useShakeDetection.ts ──> Fastify Backend (/api/sos/start) ──> playSosEmergencyAlarm()
```